Egg Segmentation and Tracking

Nosh Robotics — Computer Vision Assignment | Reproducibility and Results Report

GitHub: https://github.com/Pavani1506/nosh-robotics-cv
Tracking video: https://drive.google.com/file/d/19ZDbHlVpOppEq5EDHB9HvjZmqRPmeBTe/view?usp=sharing

1. Objective

This project explores egg segmentation in still images and individual egg tracking in a video. A pretrained Segment Anything Model (SAM) generates reference masks. Classical image-processing methods are compared against available SAM pseudo-reference masks, and a video pipeline detects eggs and associates detections across frames.

2. Dataset and project contents

The dataset contains 110 images arranged in train, valid, and test folders, with egg and Unlabeled class folders. Images are 640 × 640 pixels. It does not provide verified pixel-level egg masks. Therefore, SAM-generated masks are pseudo-reference masks, not ground truth. The Git repository contains source scripts, requirements, README, experiment records, split manifest, report, and evaluation figures. Large assets (dataset, checkpoint, generated masks, and video) are excluded and must be obtained separately.

3. Environment setup

Developed on macOS with Python 3.11.3 and an Apple M2 machine. Run the following from the repository root:

```
git clone https://github.com/Pavani1506/nosh-robotics-cv.git
cd nosh-robotics-cv
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -r requirements.txt
pip install git+https://github.com/facebookresearch/segment-anything.git
```

Download the Roboflow dataset (link in README) and place/extract it under data/egg.v1i.folder/. Download the SAM ViT-B checkpoint from the official link in README and save it as models/sam_vit_b_01ec64.pth. Keep folder names and paths consistent with the scripts.

4. Experiment execution

Inspect dataset

```
python src/inspect_dataset.py
```

Generate SAM masks

```
python src/run_sam_all.py
```

Run classical baselines

```
python src/classical_baseline.py
python src/grabcut_baseline.py
python src/run_classical_experiments.py
```

Run SAM-guided multi-egg GrabCut

```
python src/multi_egg_grabcut.py
python src/run_multi_egg_grabcut_batch_v2.py
```

Create split

```
python src/create_data_split.py
```

Generate selected SAM pseudo-reference masks

```
python src/automated_egg_evaluation.py
```

Run holdout predictions and metrics

```
python src/run_holdout_grabcut.py
python src/calculate_holdout_metrics.py
```

Create evaluation plots

```
python src/create_evaluation_plots.py
```

Run video tracking

```
python src/track_eggs_video_v2.py
```

For video tracking, place the assigned video at data/video/egg_video.mp4. Tracking outputs are written under outputs/video_tracking/v2/. Refer to README for detailed output locations.

5. Segmentation methods and experiment record

Ten trials are recorded in the experiment log. Implemented approaches include Otsu thresholding, HSV thresholding with morphology, GrabCut, and SAM-guided GrabCut. The log records parameters, observations, limitations, and rationale, including less successful attempts. SAM configuration: points_per_side=8, pred_iou_thresh=0.86, stability_score_thresh=0.90, crop_n_layers=0, min_mask_region_area=100; MPS was used where available.

6. Evaluation and results

The split manifest contains 88 development images and 22 evaluation images. The split is provisional: it was created after some experimentation and has not been independently verified for near-duplicate leakage. SAM pseudo-reference coverage was available for 14 of the 22 evaluation images; eight had no matching pseudo-reference mask.

Across the 14 matched images, mean IoU was 0.9573 and mean Dice was 0.9767 when comparing classical holdout predictions with selected SAM pseudo-reference masks. These values measure agreement with SAM-generated masks; they are not verified ground-truth segmentation accuracy.

Per-image metric and failure-case comparison figures are included in reports/figures/.

7. Video tracking

The video pipeline uses HSV-based bright/white region segmentation, geometric filtering, and centroid-distance association across consecutive frames. Configuration: ROI begins at approximately 39% of frame height; area range 6,000–95,000 pixels; aspect ratio 0.65–1.65; circularity threshold 0.48; maximum centroid matching distance 90 pixels; up to 10 missed frames. The run produced 1,386 detections and 25 track IDs. The public Drive link above contains the tracking output; the CSV records detections.

8. Limitations

- SAM reference masks are pseudo-labels, not manually verified ground truth.
- Only 14 of 22 evaluation images had matching pseudo-reference masks for the reported metrics.
- The split is provisional and has not been proven free of near-duplicate leakage.
- Thresholding and tracking may be affected by lighting, reflections, occlusion, touching eggs, and similar-looking objects.
- Reproduction requires separately downloading the dataset, SAM checkpoint, and input video; these assets are not stored in Git.

9. Links

Public GitHub: https://github.com/Pavani1506/nosh-robotics-cv
Public tracking video: https://drive.google.com/file/d/19ZDbHlVpOppEq5EDHB9HvjZmqRPmeBTe/view?usp=sharing